



Programmierpraktikum 2026



Universität Regensburg

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Chair of Programming and Software Engineering
Faculty of Informatics and Data Science (FIDS)

Regensburg, 22.04.2026

Ablauf & Orga

- Aufteilung in 2-3 Teams
 - Jeweils mind. 3 Personen
 - Ideal: 2 Teams mit je 4-6 Personen
- Betreuung
 - Judith Michael & Michael Mittermaier
- Themen:
 - Digitaler Zwilling einer Produktionsanlage
 - Digitaler Zwilling von Lernenden und Lehrenden
- Technologie
 - MontiGem Code Generator

Digital Twin for Manufacturing



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Zielsetzung

- Erstellen Sie einen digitalen Zwilling einer Produktionslinie (Fischertechnik)
- Der digitale Zwilling sollte zumindest
 - Den aktuellen Zustand beobachten können (Monitoring)
 - Ein Szenario simulieren können (Simulation)
 - Und ein weiteres Szenario ihrer Wahl realisieren
- Sie können
 - weitere „virtuelle“ Sensoren erfinden

Fischertechnik Factory



recorded data



PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEncoderCounter	{ "value": 1770, "timestamp": "2025-05-12T11:59:13.992112+00:00Z" }
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data analysis



replay, simulation,
optimization

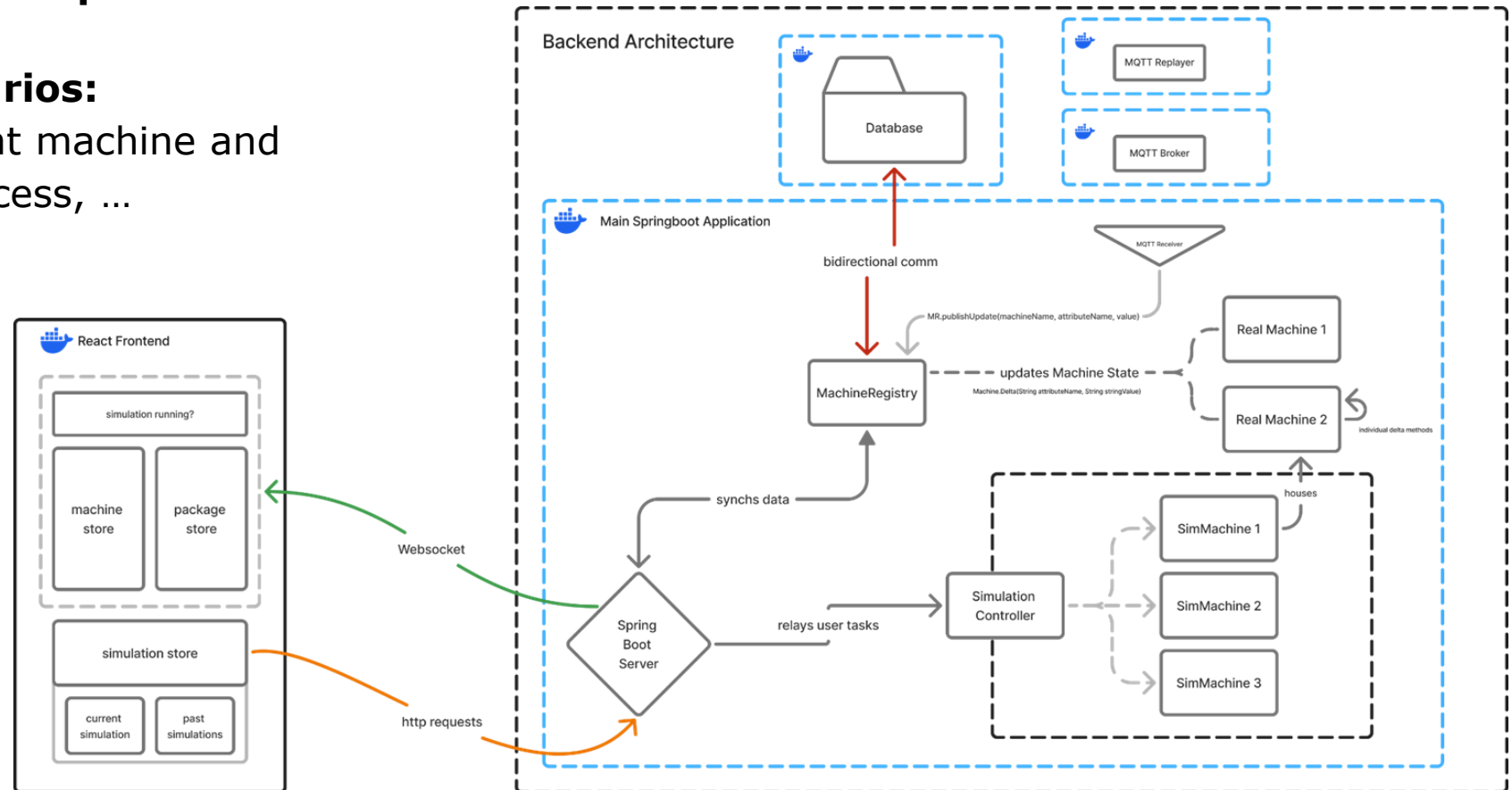


- SensTurntablePosVacuum set to "true" -> Gripper arrived at the turntable

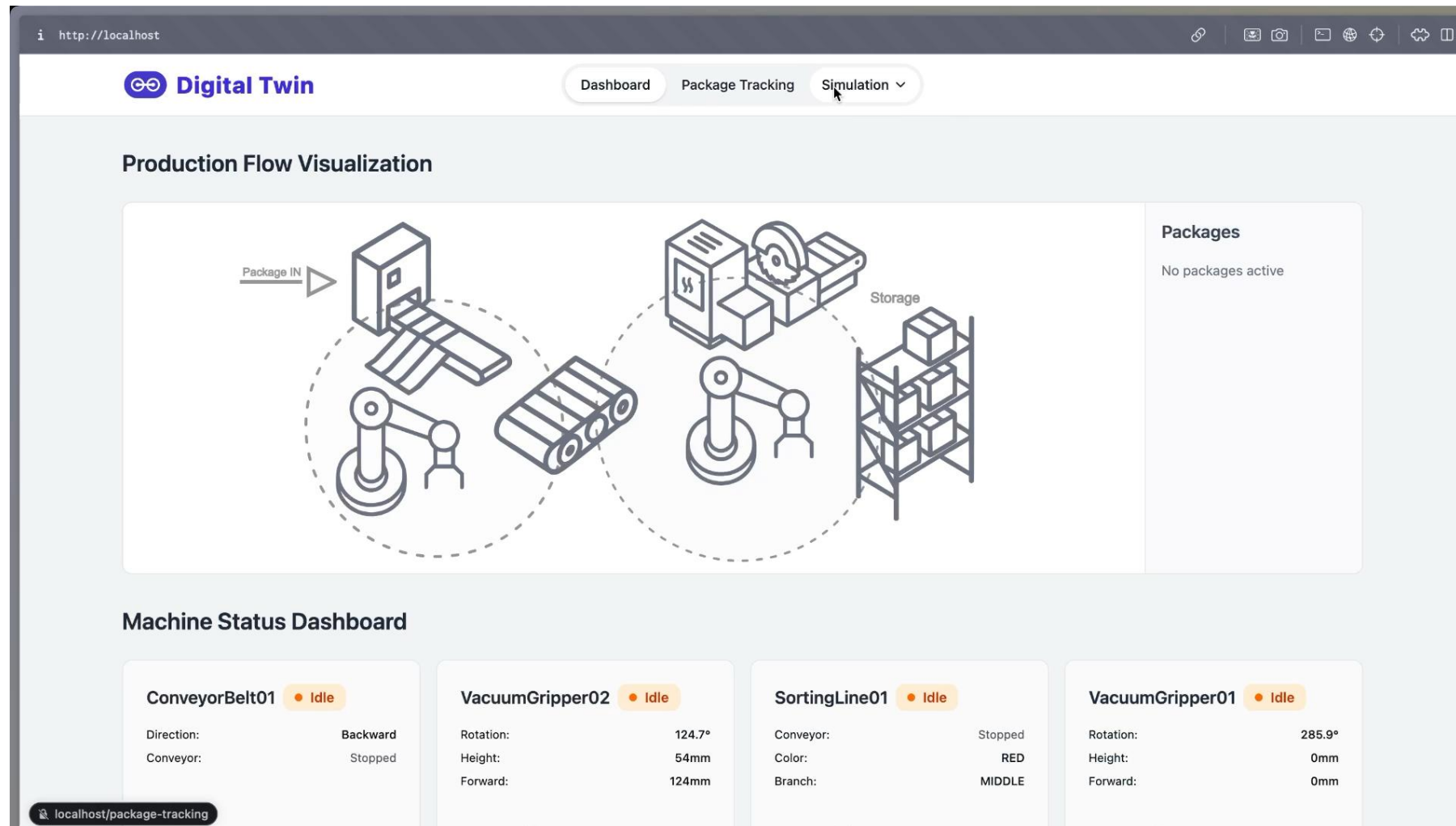
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Use Cases & Architecture

- **Monitoring the status of the production line**
- **Simulating various scenarios:**
 - E.g., loss of a redundant machine and recalibration of the process, ...
- **Analysis**
 - E.g., energy usage, errors, throughput,...
- **Optimization**
 - E.g., energy usage, speed, orders,



Simulation

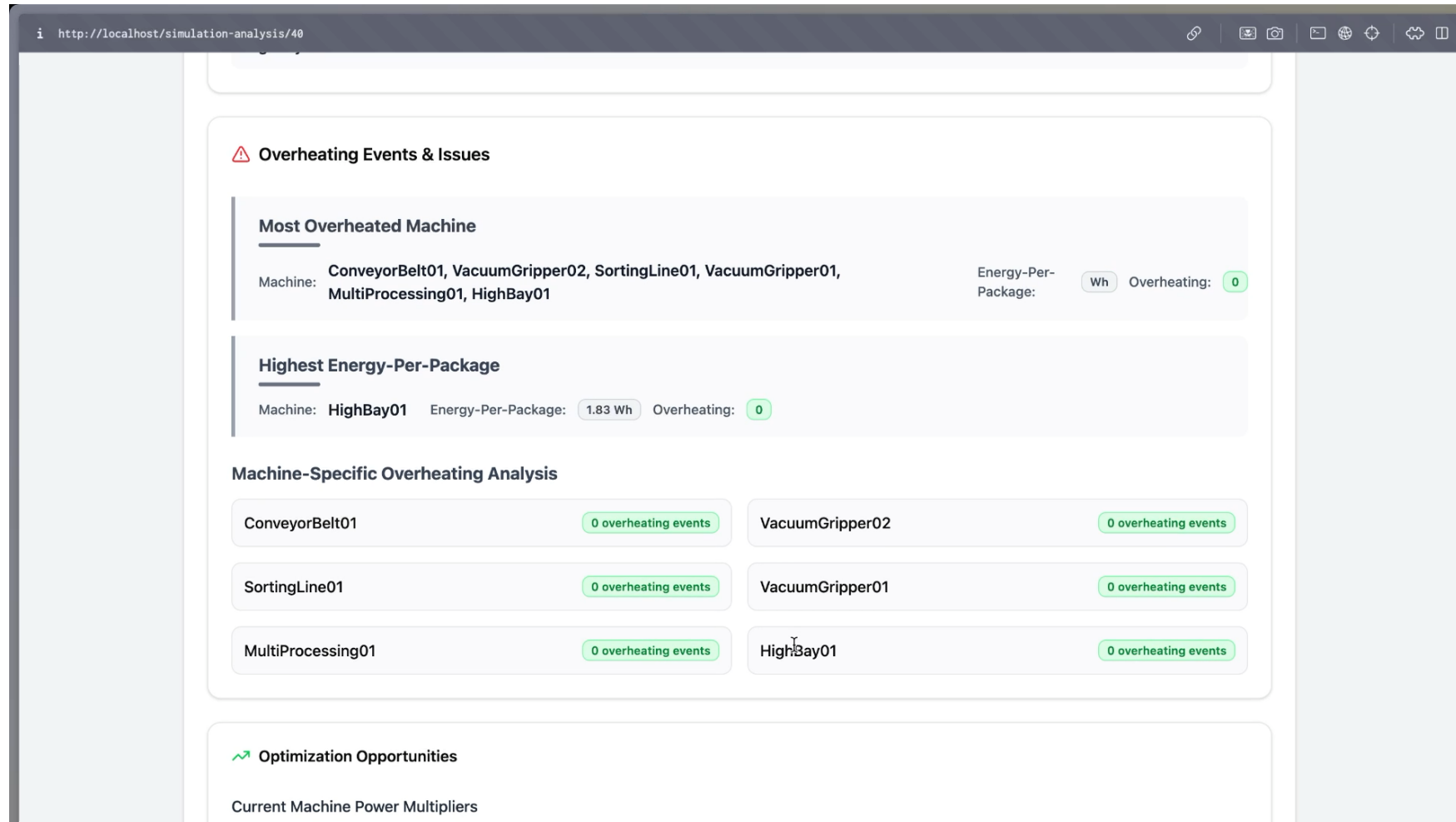


Simulation Analysis

The screenshot shows a web application for controlling a digital twin simulation. The browser address bar displays `http://localhost/simulation-control`. The application has a navigation bar with 'Dashboard', 'Package Tracking', and 'Simulation' (selected). The main interface includes:

- Start Simulation** (green button) and **Stop Simulation** (white button) controls.
- Current Configuration** section with a **Save Preset** button and a **Reset** button. It displays: Sim: 80min (0d 1h 20m), Duration: 3m 1s, 26.5x (Extremely Fast), and 1.6min/pkg.
- Simulation Status** section, currently empty.
- Basic Parameters** section (expandable) for core simulation timing and behavior settings:
 - Simulation Time to Simulate:** Input fields for Days (0), Hours (1), and Minutes (20). Below the fields, it shows 'Total: 80 minutes | Real duration: 3m 1s'.
 - Simulation Speed:** A slider ranging from 1x (Real-time) to 500x (Fast), currently set at 26.5x (Extremely Fast).

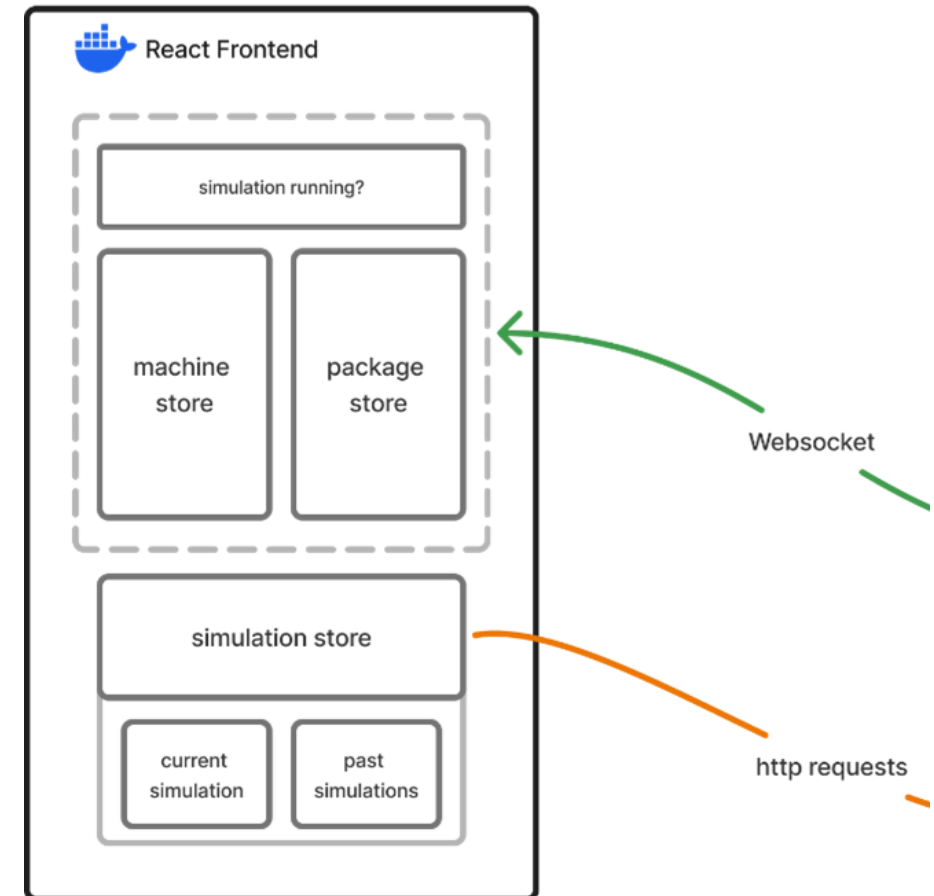
Optimization



Frontend

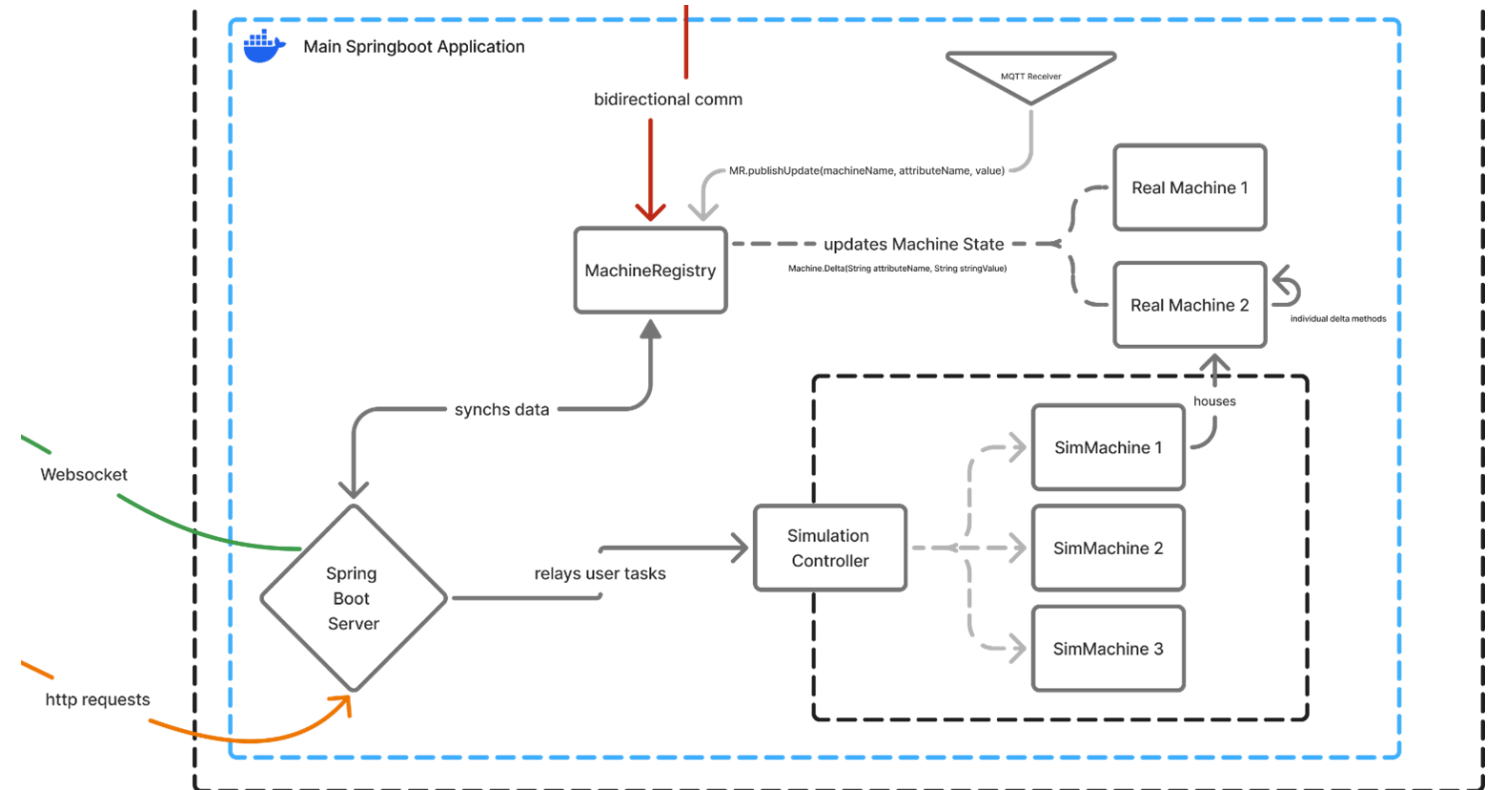
- React
 - Tailwind CSS for utility-first styling
 - shadcn UI for composable UI components
 - React Lucide for iconography
 - Zustand for state management
 - ChartJs 2 for displaying data

```
codebase/frontend/  
├── public/  
├── src/  
│   ├── assets/  
│   ├── components/  
│   │   ├── ui/           // Reusable UI components (buttons, card, dialog, etc.)  
│   │   ├── simulation/   // Simulation-related components (charts, simulation control, etc.)  
│   │   └── svgs/         // SVG components for machine visualizations  
│   ├── constants/  
│   ├── hooks/  
│   ├── lib/  
│   ├── pages/           // Pages for routing (Home, PackageTracking, SimulationControl, etc.)  
│   ├── store/           // Zustand stores for state management  
│   └── types/  
└──
```



Backend

- Spring Boot
- PostgreSQL (PG Vector)
- Hibernate



Digital Twin of Educators & Learners



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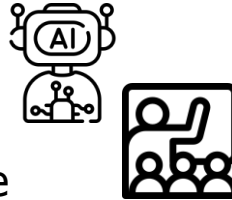
Zielsetzung

- Erstellen Sie einen digitalen Zwilling von Studierenden im Rahmen des Hochschullernprozesses
- Der digitale Zwilling sollte zumindest
 - Wichtige Informationen erfassen können (Set-up)
 - Den aktuellen Lernfortschritt beobachten können (Monitoring/Interviewing)
 - Und ein weiteres Szenario ihrer Wahl realisieren
- Sie können
 - weitere „virtuelle“ Sensoren erfinden
- Einschränkungen
 - Nur Studierendenperspektive?

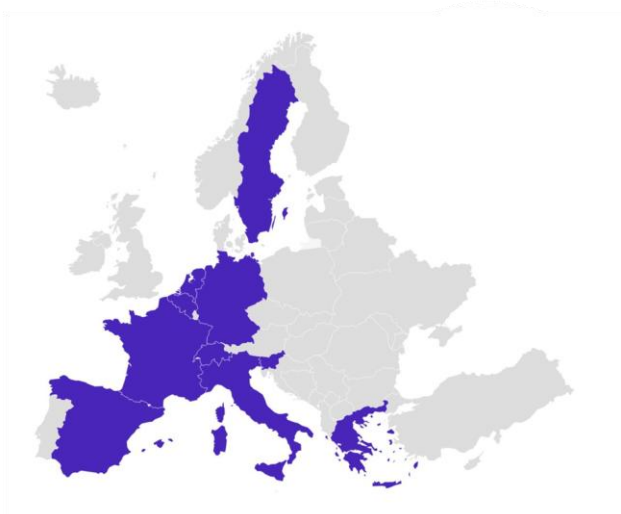
Digital Twins for Learners and Educators

Part of the Erasmus+ AIM-PRO project: AI Literacy for Multidisciplinary Professional Readiness and Outreach

- *AI literacy* becomes a **fundamental competency**: in higher education, vocational training, and for the workforce
- *Project aims*: developing innovative, multidisciplinary **educational frameworks** that integrate AI literacy with Open Education, Generative AI, Model-Driven Engineering and Digital Twins
- Our SE-UR focus:
 - *Creating digital twins for learners and educators*, e.g., to prepare educational material fitting to different preferences and skills, track learning progress,...
 - Tailor learning material to different profiles and skills



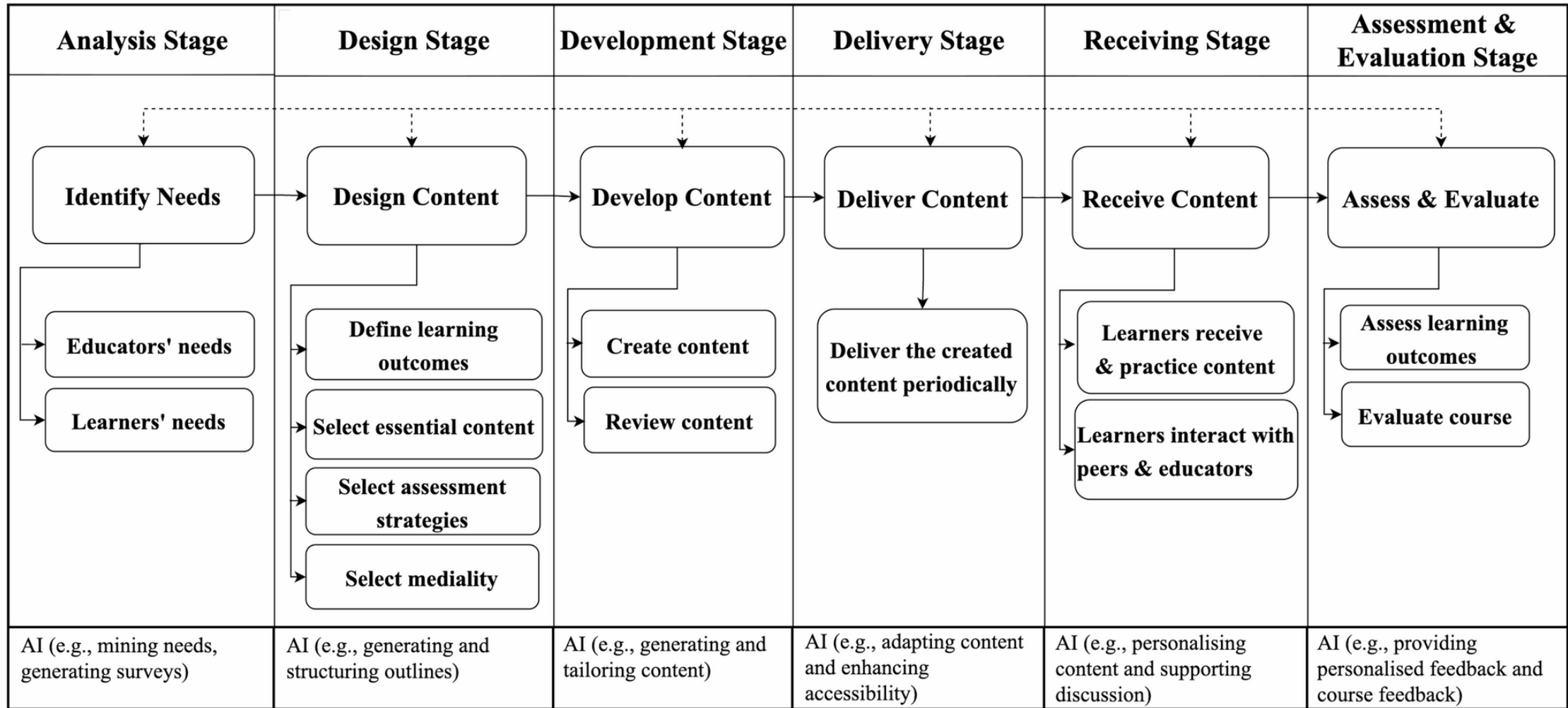
AIM-PRO Consortium



**Co-funded by
the European Union**

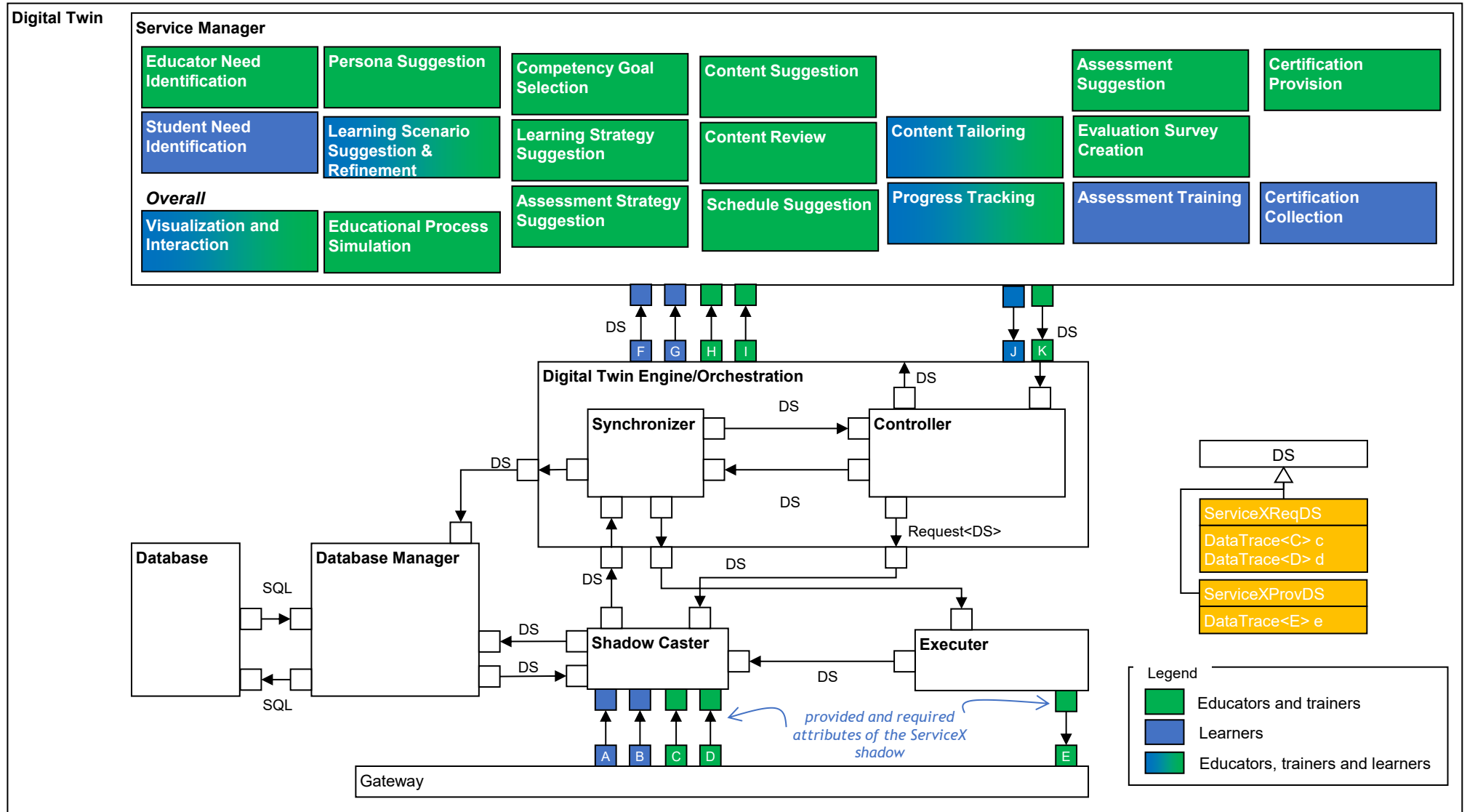
Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Education and Culture Executive Agency (EACEA). Neither the European Union nor EACEA can be held responsible for them.

Based on: Microlearning AI-Integrated Instructional Design (MIND) Model to support effective instructional design to enhance learning outcomes



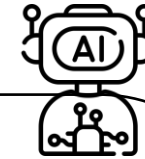
<https://doi.org/10.1038/s41598-025-24910-y>

Core DT Architecture (Concept)



Digital Twins for Learners and Educators

Some possible AI agents



Adaptive Learning Path Personalization Agent

Input:

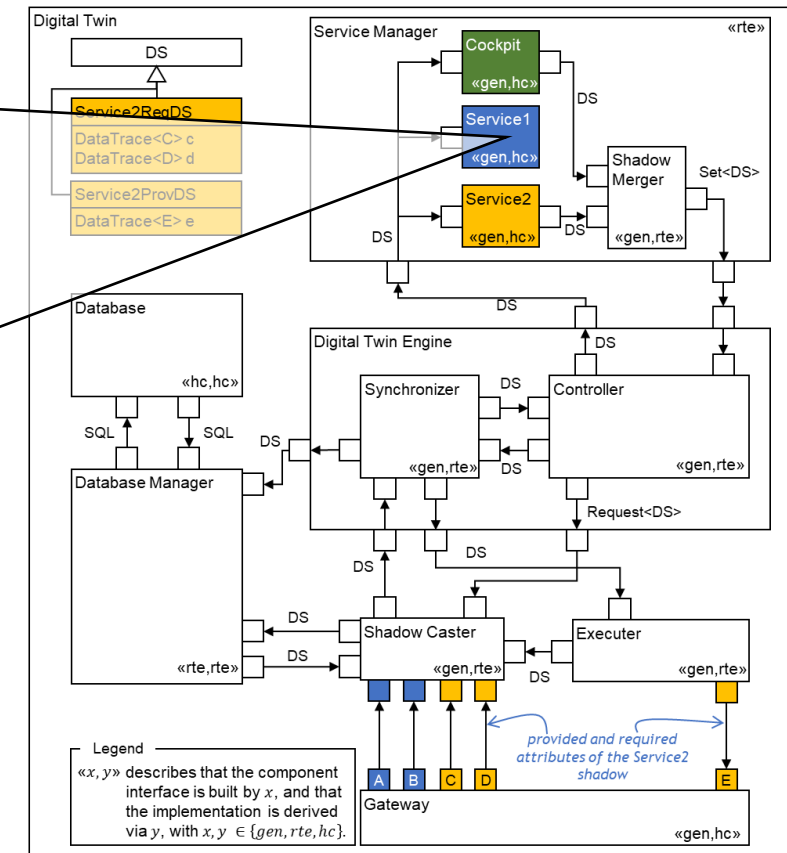
- Learner profile (background, level, preferences)
- Historical performance
- Current learning objectives

Functionality:

- Generate personalized learning paths
- Dynamically suggest remedial or advanced activities
- Adjust pacing, modality, and difficulty without altering objectives
- Provide real-time formative feedback and next-step recommendations

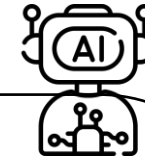
Output:

- Personalized path variant
- Ranked list of recommended activities or resources
- Explanation of adaptation decisions (for transparency)



Digital Twins for Learners and Educators

Some possible AI agents



Content Customization Agent

Input:

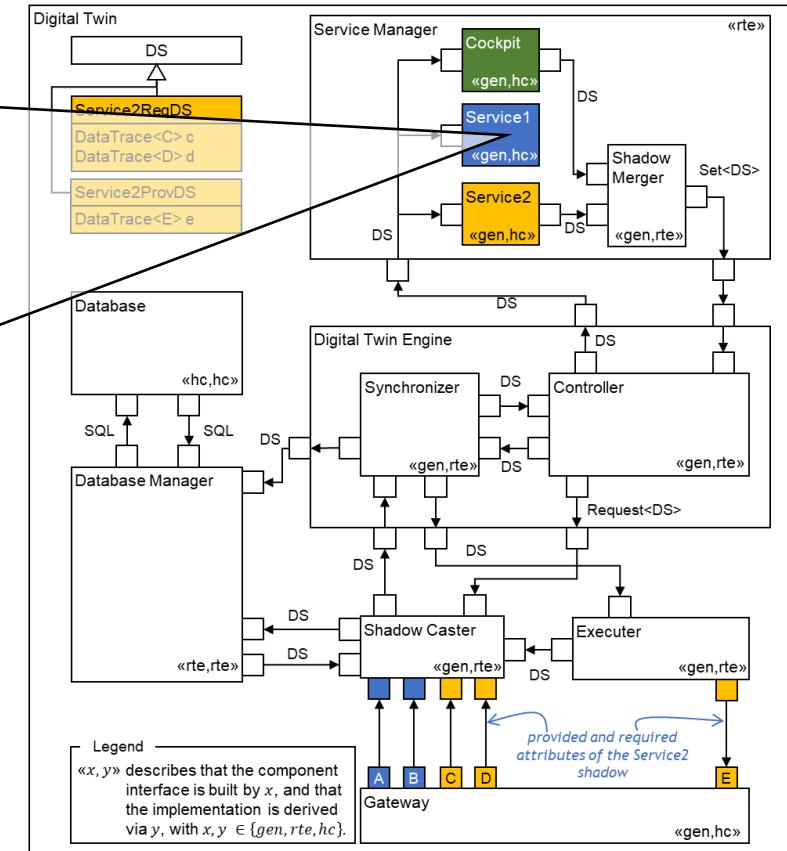
- Base content (activity, assessment, module)
- Learner profile and accessibility requirements
- Delivery context (HE, SME, multilingual setting)

Functionality:

- Adapt language level, modality, and complexity
- Generate multilingual variants
- Provide accessible formats (simplified text, structured steps)
- Preserve learning objectives while modifying presentation

Output:

- Context-adapted content variants
- Accessibility-compliant materials
- Summary of modifications relative to the base version



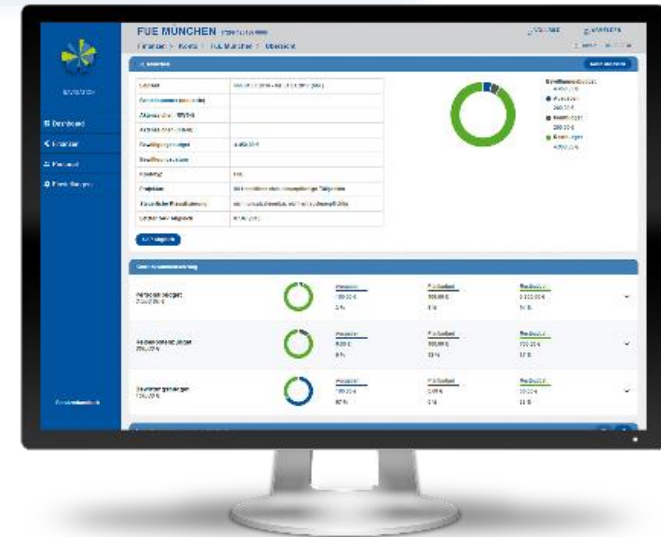
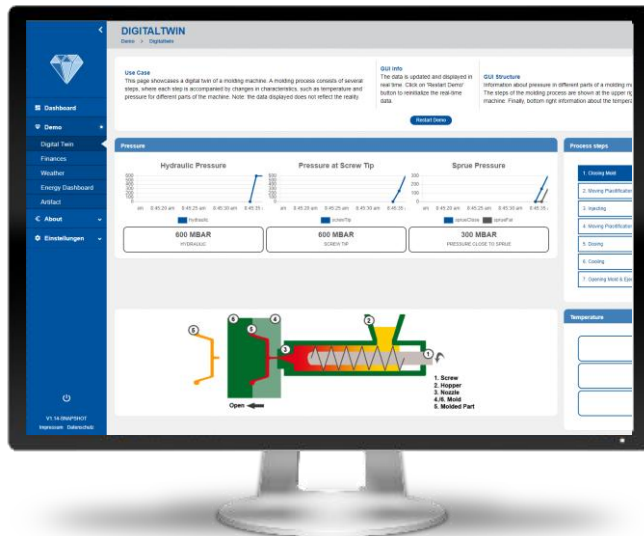
MontiGem



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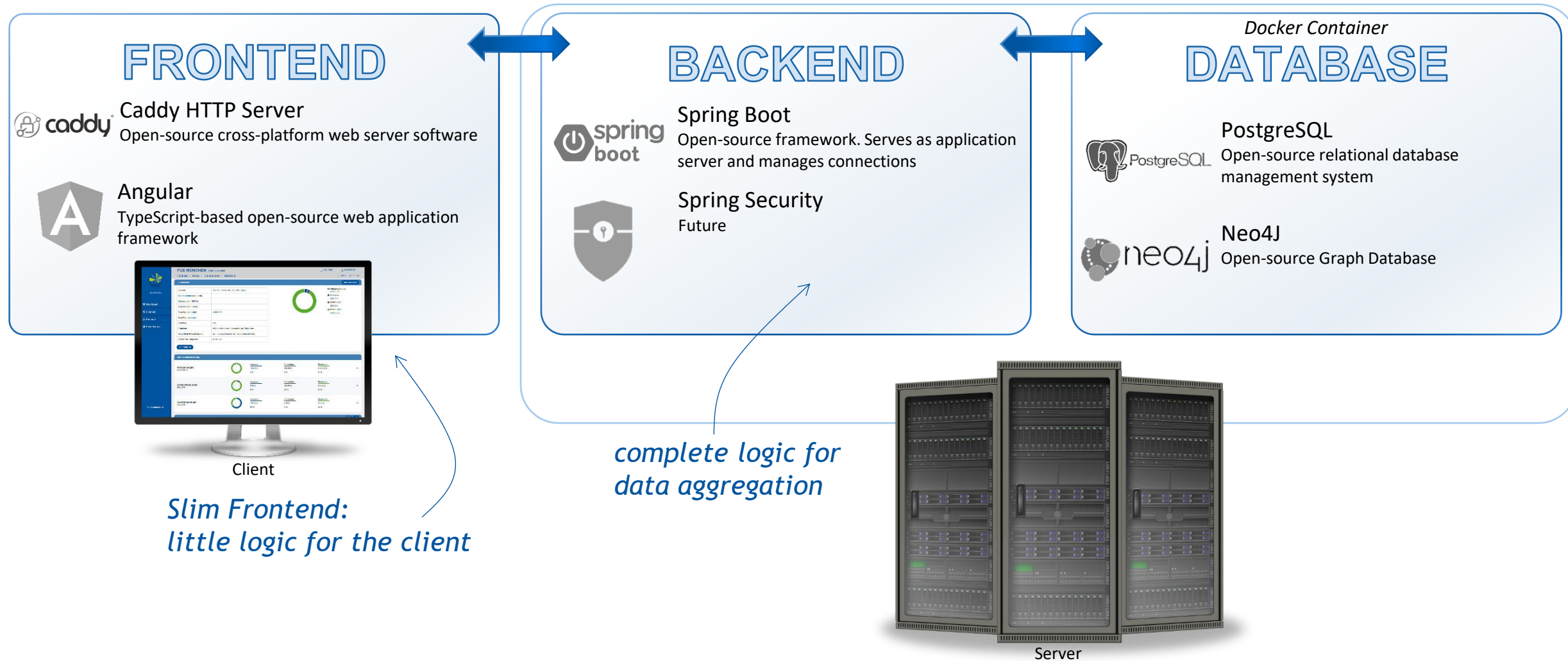
MontiGem 3

MontiGEM is a [framework](#) that provides support to generate large parts of a [web application](#) for a domain specific Enterprise Information System consisting of [data base, persistence layer, command infrastructure](#) and [graphical user interface](#) based on the [input](#) of [models](#).



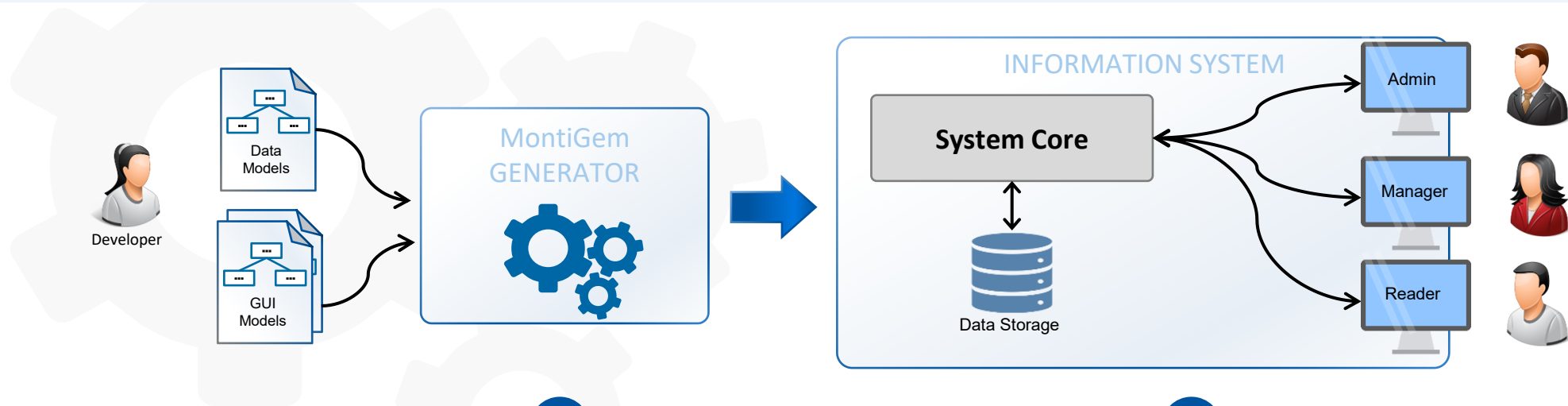
<https://monticore.pages.rwth-aachen.de/umlp/index.html>

Architecture of the (Generated) Enterprise Information System



MontiGem 3: Model-Driven Software Engineering of Information Systems

The **MontiGem** framework provides support to generate large parts of a **web-based domain-specific information system** consisting of **data base, persistence layer, command infrastructure** and **graphical user interfaces** based on the **input in models**.



1 Define models

- Required: Class Diagrams (Data structure, subsets for views), GUI Models (Views)
- Optional: OCL (Input data validation), BPMN (Workflows)
- CD2GUI: only Class Diagrams (Data structure)

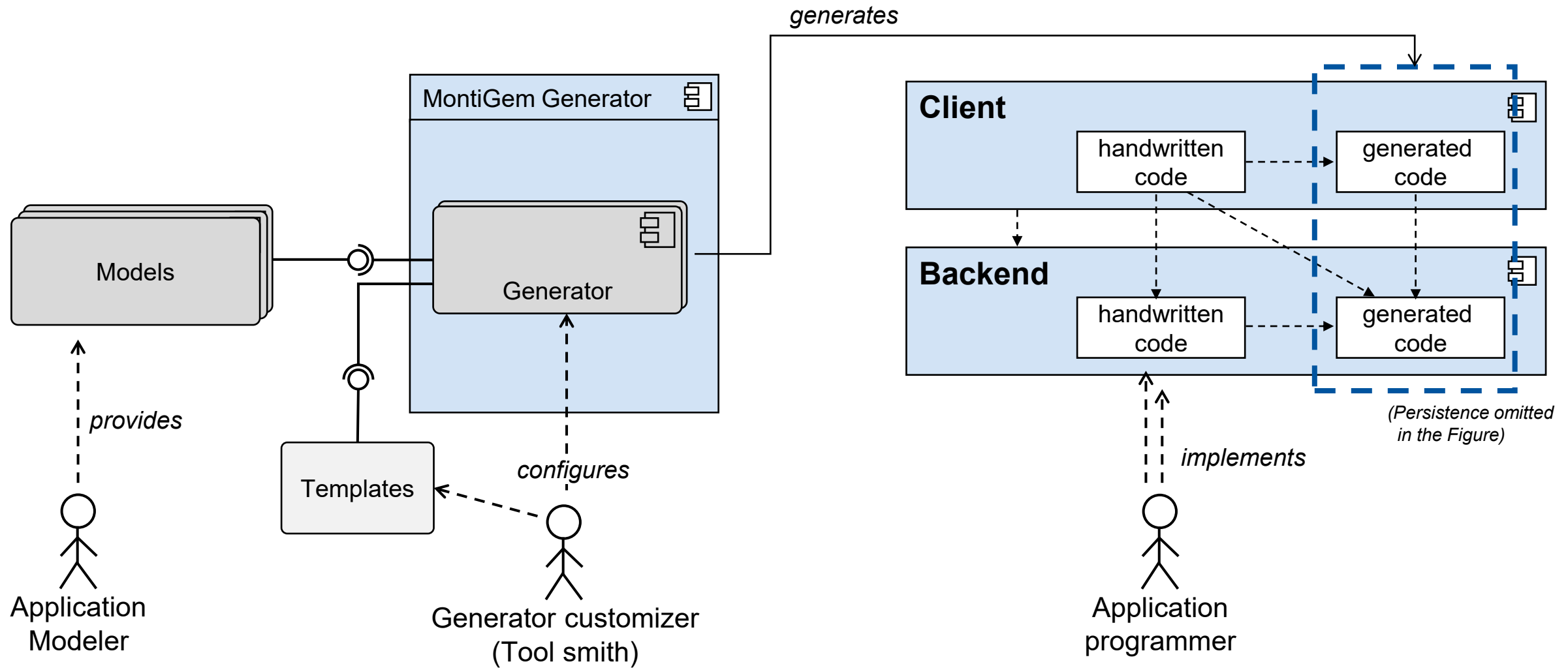
2 Generation of the „System Core“

- Database
- Communication Infrastructure between Frontend and Backend
- Graphical User Interfaces

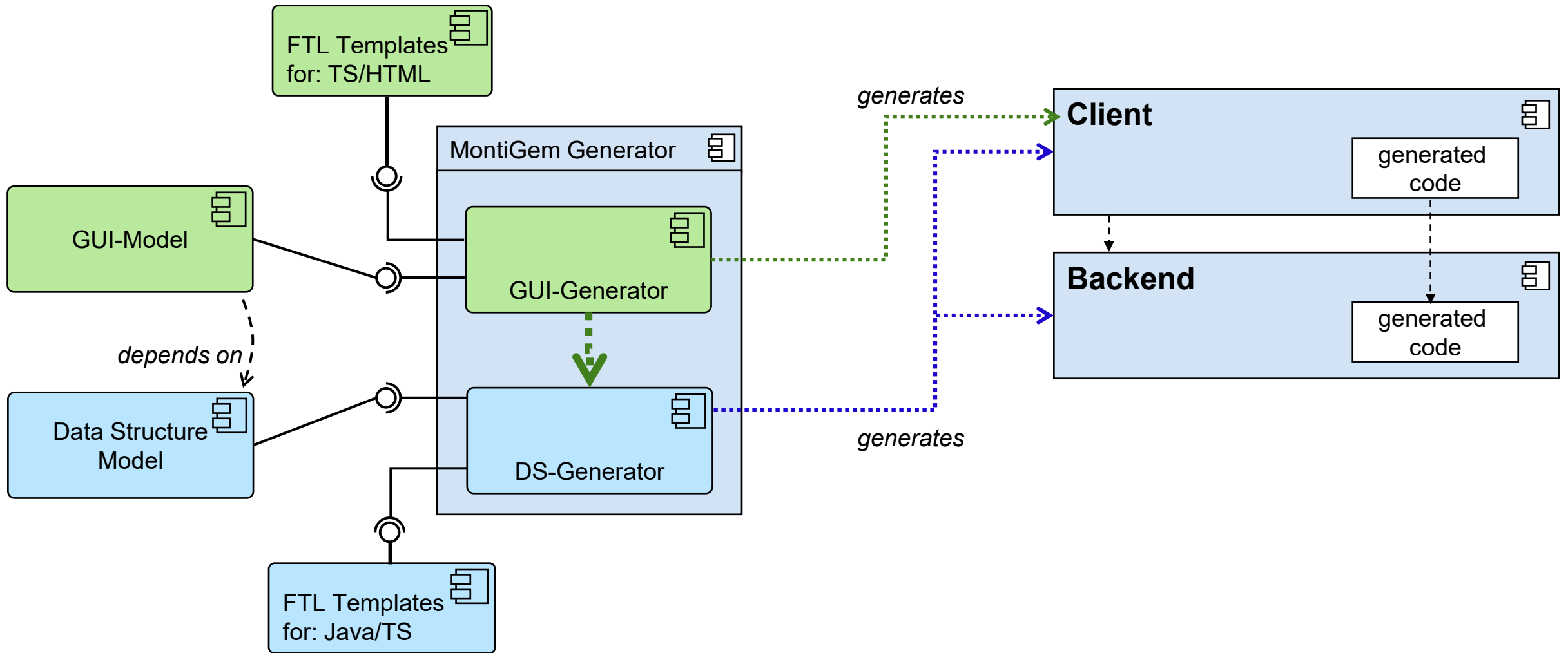
3 Hand-Written Extensions

- Allow for re-generation
- Customize Appearance
- Add business logic

Generator Architecture | Principle



Generator Architecture | Details



Communication Architecture and its generated Parts

